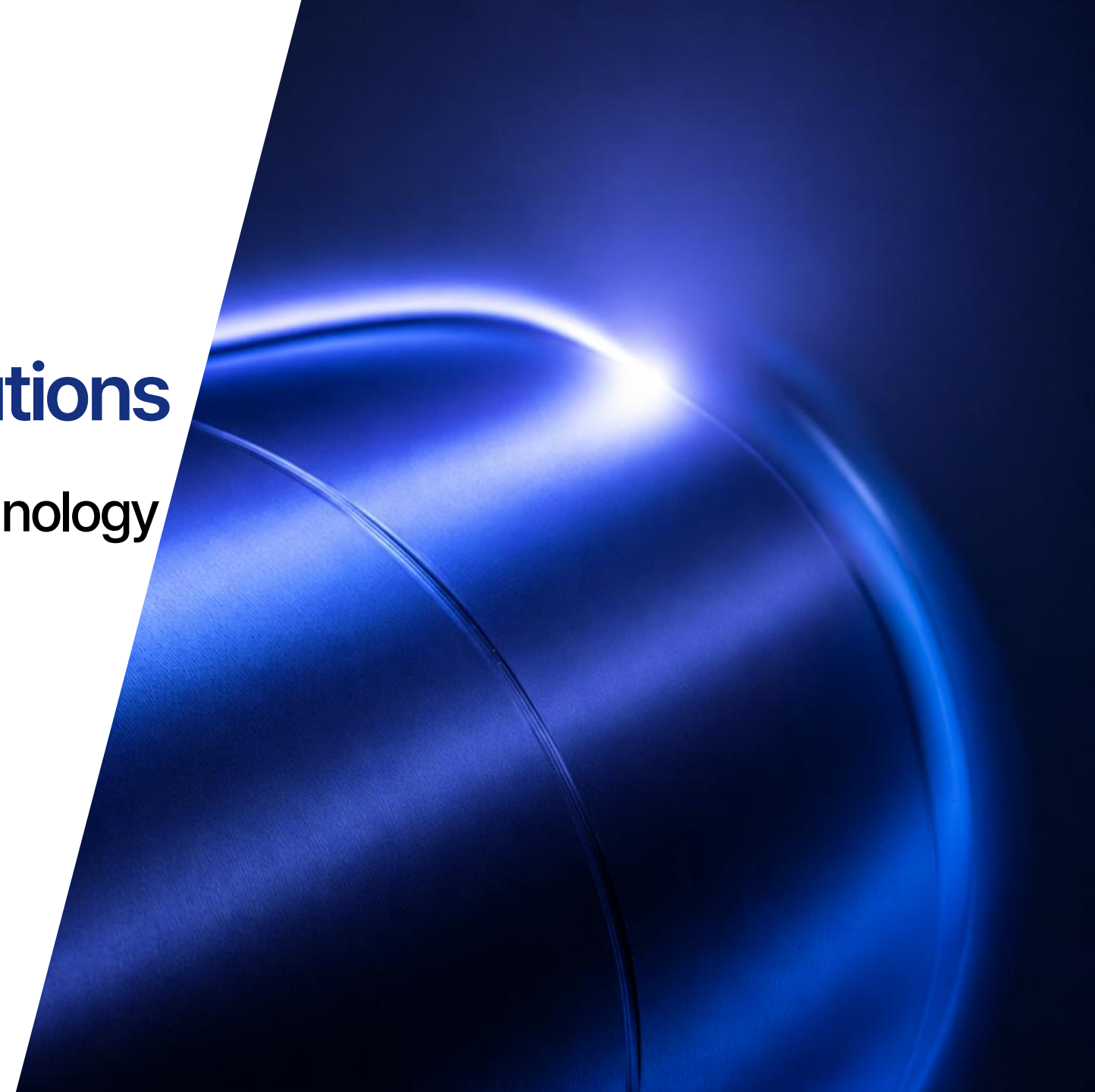
CONFIDENTIAL

Customized
Surface Treatment Solutions

World-Class Plasma Coating Technology

Innovative Plasma Coating for a Sustainable Future



INNOVATIVE PLASMA COATING
INNOTION TECH

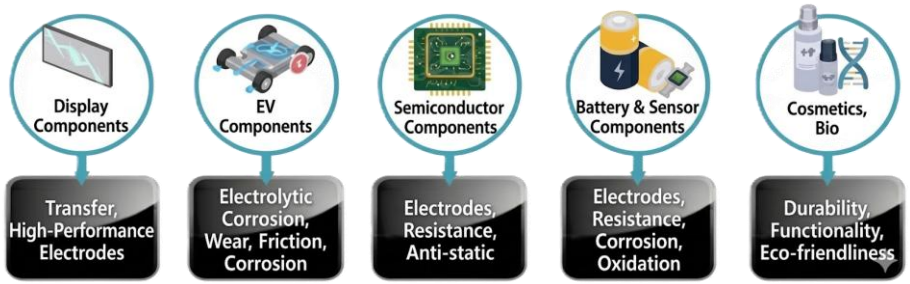
CONFIDENTIAL

Surface treatment Total Solution Provider, INNOTIONTECH

CEO	Jeong Ki Hong
Established	June 26, 2019
Capital	USD 623,800
Employees	54
Industry	Manufacturing (Plasma coating and metal surface treatment)
Main Products	Plasma coating systems, materials, and coating services
Main Products	22, Maewhasandan 3-gil, Siheung-si, Gyeonggi-do, Korea
Busan Plant	13, Dasan-ro, Saha-gu, Busan, Korea
U.S. Subsidiary	401 Three Springs Drive, Suite D, Weirton, WV 26062, USA
EU Subsidiary	CTPark Poruba, 17. listopadu 6243, 708 00 Ostrava, Czechia (Incorporation Scheduled for September)
Website	https://innotiontech.co.kr

InnotionTech

An R&D-driven specialist in eco-friendly plasma coating,
we provide integrated "System, Material, and Service" solutions
to enhance performance and durability.
We pursue the goal of becoming a global leader in
"Materials, Components, and Equipment," delivering customer
value in the rapidly evolving future industrial market.



CONFIDENTIAL

The Journey of Innotiontech

Innotiontech Co., Ltd.

INNOVATIVE PLASMA COATING
INNOTION TECH

Unrivaled Technological Competitiveness

Providing customized solutions tailored to specific customer requirements.

Proven technological reliability through joint development and application cases with global leading companies.

Growth and Scalability

Positioned as a key process/material company in the rapidly growing secondary battery, EV, and semiconductor markets.

Proven technological reliability through joint development and application cases with global leading companies.

Leading Global Companies with Us

Global Top-tier Partners
: Battery, IT, Automotive, Semiconductor

2026
- Establishing a European corporate entity in the Czech Republic (Corporate registration to be completed in Sep.)
- Completed validation of surface treatment for dry electrode equipment
- Successfully commercialized vision inspection systems
- Purchased a new plant in Busan
- Selected for the Unicorn Bridge project hosted by the Ministry of SMEs and Startups
- Conducting advanced R&D with global top-tier electronics, display, and steel conglomerates

2025
- Established InnotionTech US Subsidiary (Weirton, West Virginia, USA)
- Awarded main press roll coating contract for Samsung SDI's solid-state battery development pilot line

2024
- Successfully developed large-scale hybrid plasma coating system equipment and process technology
- Established Busan plant

2023
- Selected as a "Super Gap 1000+" company by the MSS
- Initiated development of large-scale equipment for secondary batteries

2021
- Signed NDA with Samsung Electronics
- Registered as a vendor for ServeOne (LG Group affiliate)
- Completed the Materials, Parts & Equipment Startup 100 Program

2020
- Signed NDA with LG Electronics

2019
- Established Innosion Tech Co., Ltd.

CONFIDENTIAL

Patents Underpinning Our Core Technologies

Core & Foundational Technology Patents

1 Nanocomposite Coating for Rollers in Battery Electrode Manufacturing Equipment



Core Surface Treatment Technology for Battery Rolls

Roll Coating Technology for Dry & Wet Electrode Manufacturing Lines

Korea Patent | U.S. Granted | Europe & Asia

2 Amorphous Carbon Coating Method for Rubber Rollers



Surface Treatment Technology for Rubber Rollers and Specialized Materials

Functional Coating Solutions for Specialized Materials Core Surface TreatmentTechnology for Rubber Rollers

Korea Patent | PCT | U.S. Application

3 High-Density, Large-Area Metal Coating & Equipment



Metal Coating Technology Replacing HCr and WC Underlayers

Alternative to HCr Electroplating & WC Thermal Spray ESG-Compliant Solution

Korea Granted | U.S. & Europe Application

Korea Patents

18 Patnents | 14 Granted / 4 Pending

Overseas Patents & PCT Applications

13 Patnents | PCT 3 / U.S. 5 / Europe 3 / Japan 1 / China 1

Korean Trademarks

4 Patnents | 4 Resistered

Patents Underpinning Our Core Technologies

IP Portfolio Overview

Global Patent Network

18

Patents

14 Granted / 4 Pending

Overseas Patents & PCT

13

Patents

PCT 3 / U.S. 5 / Europe 3 / Japan 1 / China 1

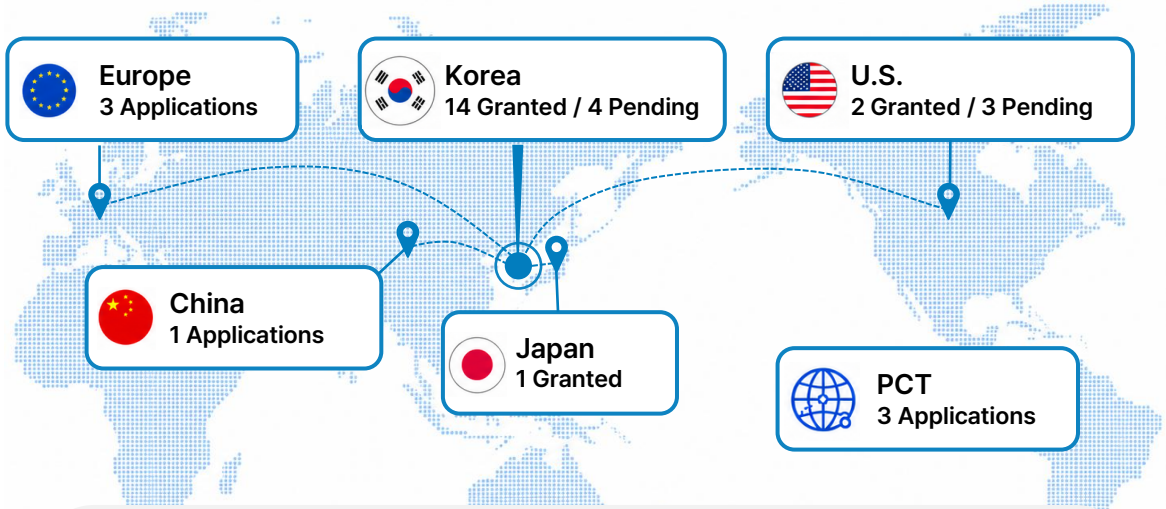
Korean Trademarks

4

Trademarks

4 Registered

Global Patent Network



Building a Global IP Portfolio by Expanding Core Technologies to the U.S., Europe, Japan, and China

Additional Domestic Patent Portfolio

Secondary Battery
Electrode Manufacturing Equipment
Roll Surface Treatment

Future Mobility
Hybrid Vehicles
EV Drivetrain Components

Display
Functional Coatings for Substrates

Industrial Materials & Components
Rubber Rollers & Packaging Materials
Surface Treatment

Manufacturing Equipment & Processes
Coating Equipment & Manufacturing Systems

CONFIDENTIAL

INNOVATIVE PLASMA COATING INNOTION TECH

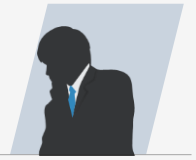
Growth Organization Led by Experts

- **CEO**
  - **Corporate Strategy & Planning**
    - Sales/Tech Strategy Enhancement Team
    - Tech/Procurement Strategy Enhancement Team
  - **Management Planning & Support**
  - **R&D Institute**
    - Material Process Team
    - System Development Team
  - **Coating Business Division**
    - Large-scale Manufacturing Team
    - Small & Medium-scale Manufacturing Team
  - **Busan Branch** (Secondary Battery Business Division)
  - **U.S. Subsidiary**



CEO
Jeong Ki Hong

- ☑ Overall Management & R&D Strategy
- ☑ 10+ Years of Experience in Plasma Coating



Executive Director
In Ki Hong

- ☑ Technical Sales & Commercialization Strategy
- ☑ 10+ Years of Experience in Plasma Coating



Managing Director
Myoung Ki Hong

- ☑ Busan / Secondary Battery Business Division
- ☑ Mass Production Setup & Operations for Press Roll Coating



Corporate Management
Kim Sung Youl

- ☑ HR, General Affairs & Internal Control
- ☑ Company-wide Security Management



Principal Engineer
Lee Soo Yong

- ☑ Lead Engineer, Advanced Coating Processes



Principal Engineer
Lee Ju Hyeong

- ☑ Lead Engineer, Equipment Development & Process Optimization



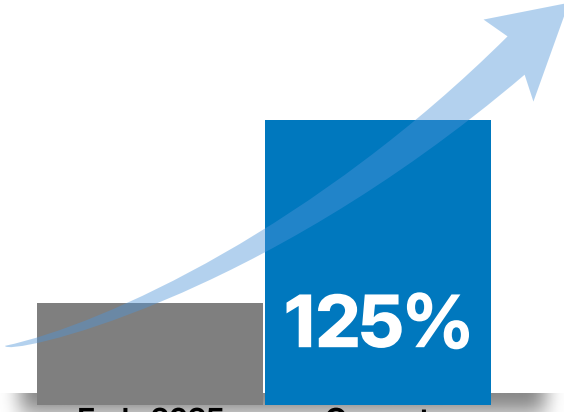
Manager
Kang Dong An

- ☑ Team Leader, Coating Business Division (HQ)
- ☑ Large-scale Equipment Setup & Customer-specific Process Implementation



Team Leader
Kim Jin Li

- ☑ Finance, Funding & Accounting
- ☑ Financial Closing, Tax & External Audit Management

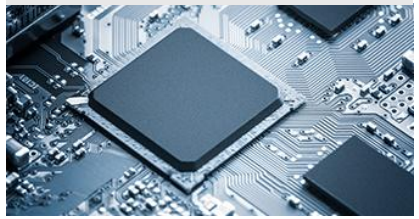


Workforce increased by 125% over the past year

- Strengthening R&D and Process Engineering Capabilities
- Expanding Talent Pool for Global Growth and IPO Readiness
- Establishing Internal Controls and Public-company Standards

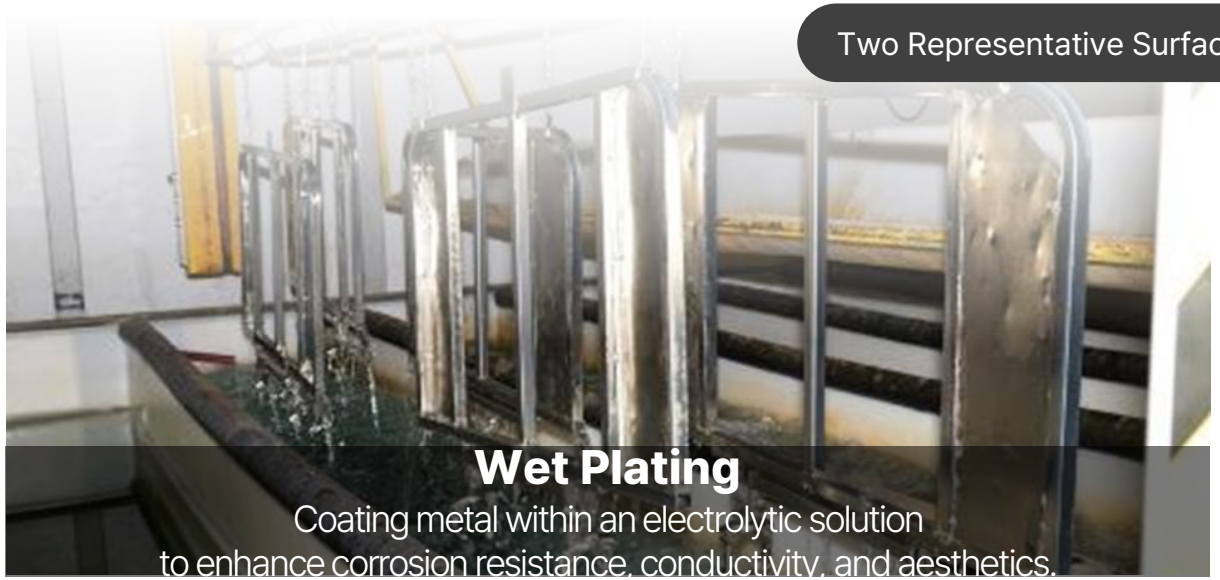
CONFIDENTIAL

Types of Surface Treatment Technology

Electronics & Semiconductors	Automotive	Aerospace	Machinery	Medical Devices / Consumer Goods
				

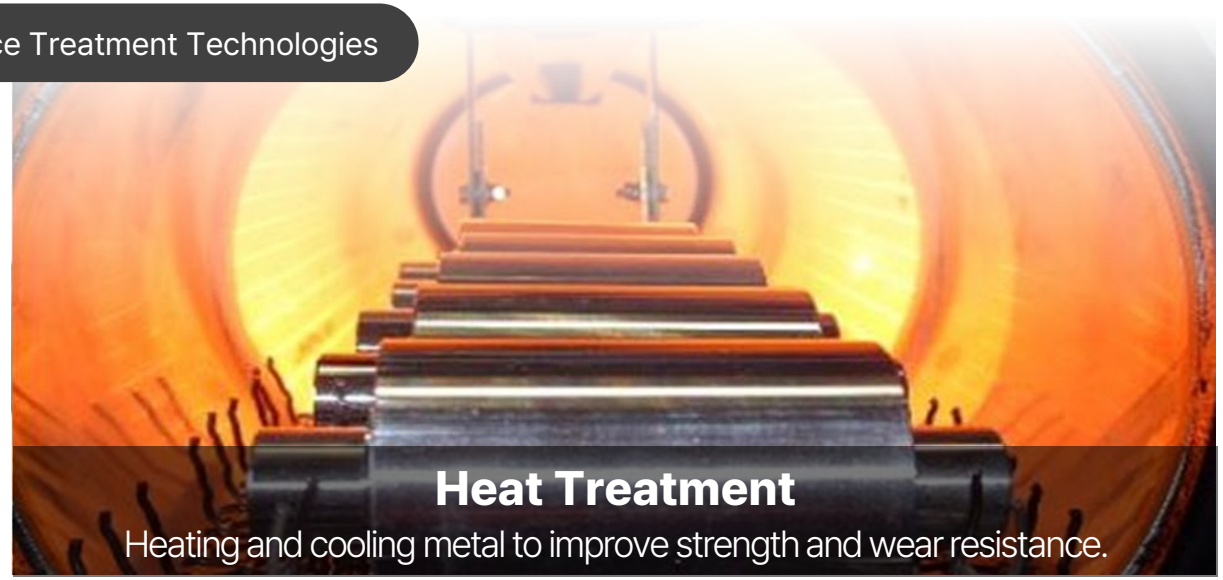
Surface treatment is an essential core technology required to strengthen product durability and reliability, and to enhance functionality and appearance.

Two Representative Surface Treatment Technologies



Wet Plating

Coating metal within an electrolytic solution to enhance corrosion resistance, conductivity, and aesthetics.

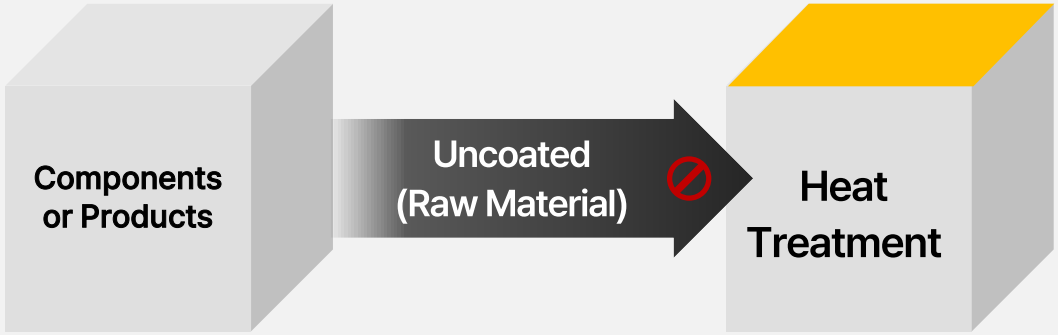


Heat Treatment

Heating and cooling metal to improve strength and wear resistance.

CONFIDENTIAL

Heat Treatment/Wet Coating Methods Falling Behind Recent Trends



Short Replacement Cycle!

Cost ↑ Maintenance ↑

Performance and Durability

Component or Product Failure Occurs



Liquid-Based Issues

Density ↓ Durability ↓ Odor ↑

Wastewater and Waste ↑

Cost (for Thick-film coating) ↑

Plasma Coating Technology

Addressing the limitations of conventional methods

Functional Film(nm~um)

Substrate

(Surface of existing components or product materials)

CONFIDENTIAL

More Eco-friendly and More Rational Technology



Conventional Heat Treatment Technology	Conventional Wet Coating Technology	Plasma Coating Technology
Improving mechanical properties by heating and cooling metal to transform its internal structure	Coating and plating in a liquid (solution) environment	Dry coating
		Surface coating using plasma in a vacuum or low-pressure state
Enhancing hardness and strength	**Effective for corrosion prevention and aesthetic improvement**	**Zero waste, Zero wastewater**
But	**But**	**Zero**
• Increasing fatigue life • Energy consumption↑ • Occurrence of oxidation and distortion	• Issues with uniformity • Environmental burden such as wastewater and waste	• Possible to realize material properties desired by customers • Precise and uniform treatment possible • Possible to shorten pre-treatment processes and reduce costs

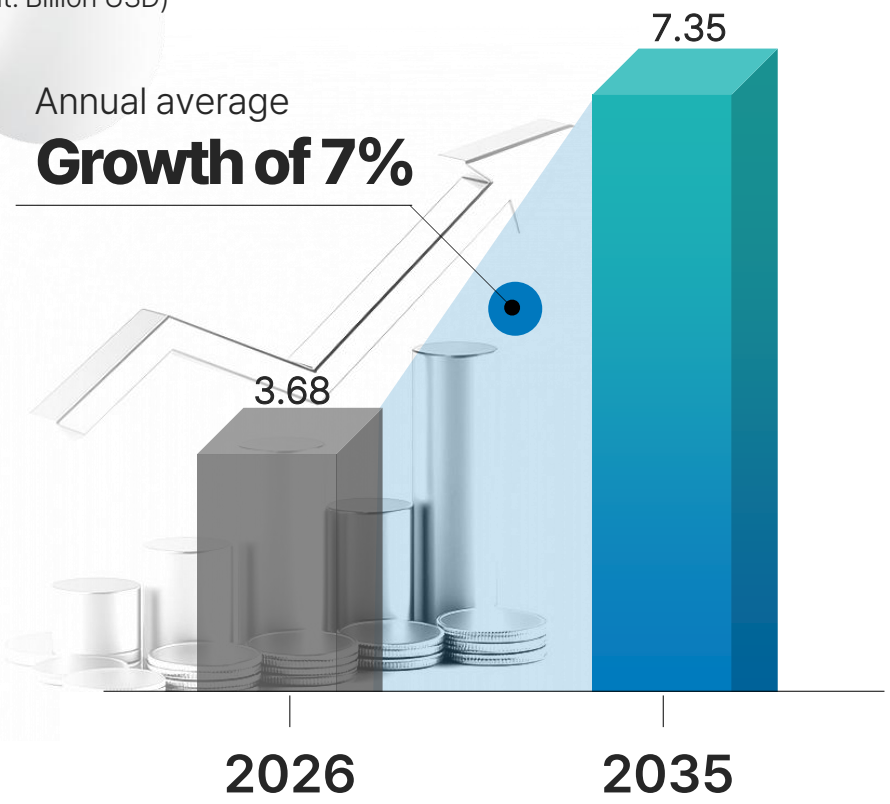
CONFIDENTIAL

INNOVATIVE PLASMA COATING INNOTION TECH

Emerging as an innovative technology across various industries

Continuously growing plasma coating market

(Unit: Billion USD)

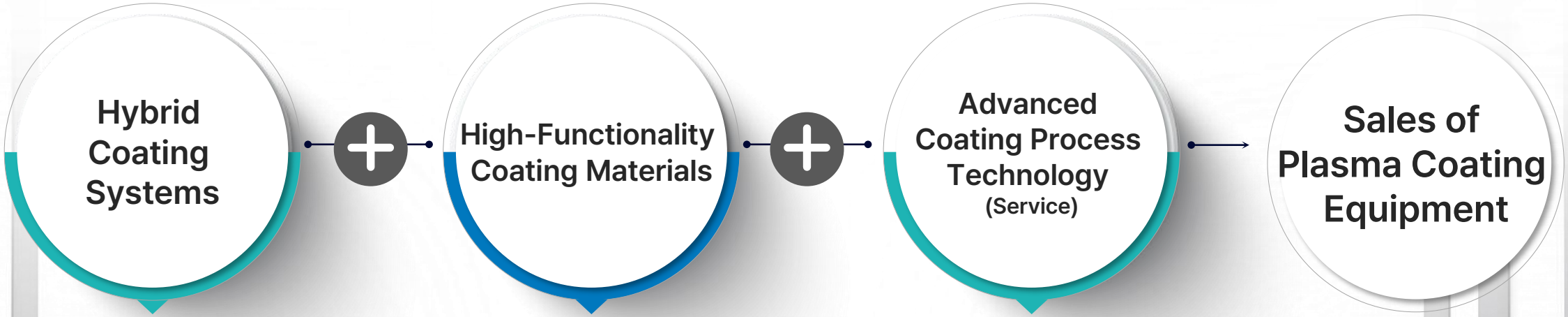


Source: Business Research Insights_'26.01.19

- Surge in demand for extending the lifespan of components in ultra-fine semiconductor processes and for insulation/heat dissipation coatings in electric vehicle (EV) battery systems
- Application of thermal barrier coatings and wear/friction reduction coatings for core aerospace components, such as jet engine turbine blades and rocket parts, exposed to extreme temperatures (above 1,500°C) and corrosive environments
- An eco-friendly dry process with lower hazardous substance emissions compared to conventional wet coating; demand for eco-friendly dry coating is expanding in line with global ESG trends and strengthening regulations

CONFIDENTIAL

Business Model Based on Total Plasma Coating Solutions



Providing Customized Comprehensive Plasma Coating Solutions

As-Is: Centered on Coating Systems + Materials + Process Services

To-Be: Expanding into a Comprehensive Platform Model including Equipment Sales

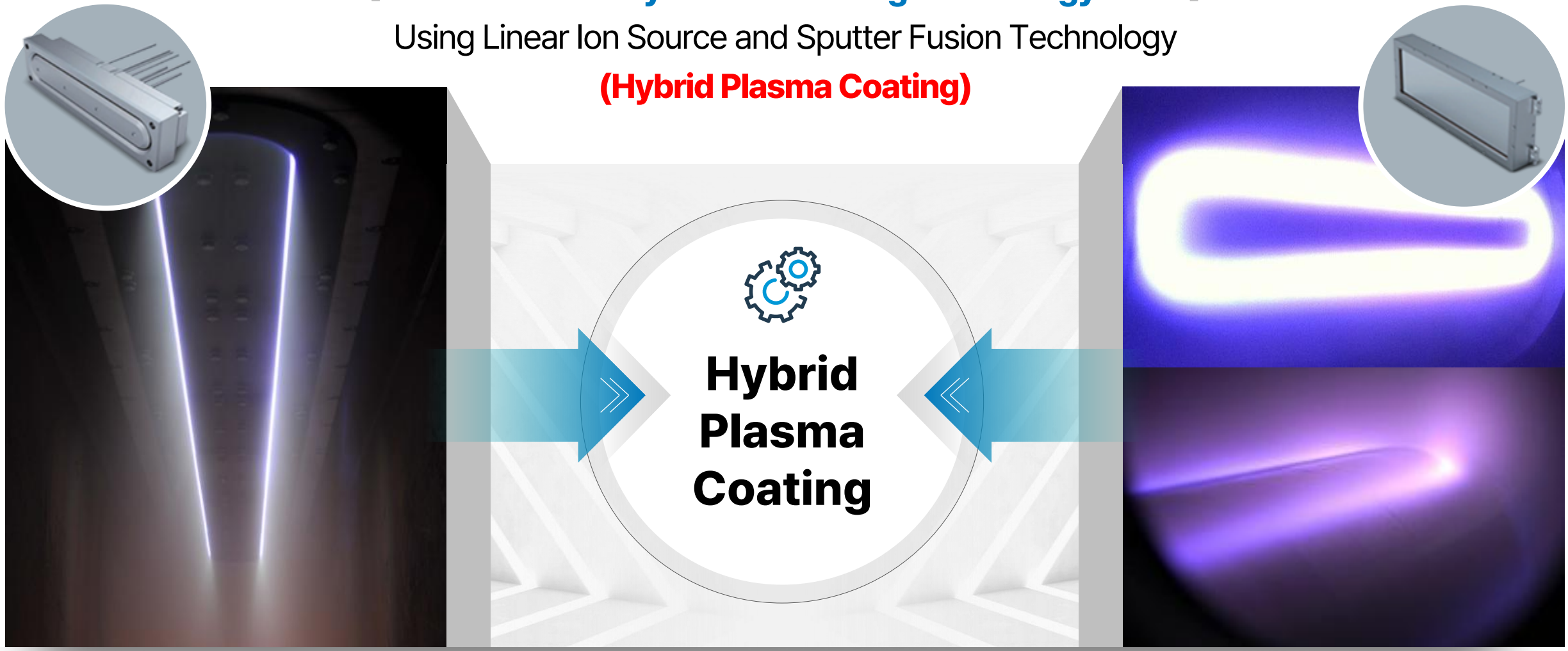
CONFIDENTIAL

The Core of Our Differentiated Plasma Coating Technology

Eco-friendly Plasma Coating Technology

Using Linear Ion Source and Sputter Fusion Technology

(Hybrid Plasma Coating)



CONFIDENTIAL

Hybrid Plasma Coating Technology Positioning

PVD Plasma Coating

Strengths

- High hardness, wear resistance, low friction
- Low-temperature process (200~500°C)
- excellent mass production stability

Limitations

- Limited coverage on complex shapes
- Constraints in realizing high-functionality thin films

Applications

- Molds, tools, equipment parts

Hybrid Plasma Coating

Fusion of PVD durability + CVD functionality

- High density and high deposition rate
- Enhanced step coverage (coverage on complex shapes)
- Low temperature (~70°C) plasma process

Overcoming PVD and CVD limitations simultaneously

CVD / PECVD Plasma Coating

Strengths

- Excellent step coverage
- Insulation barriers

Limitations

- High-temperature process (800~1,000°C) and safety concerns
- High equipment and operational costs

Applications

- Semiconductors, displays, components for high heat/chemical resistance environments

Solution for overcoming single-process limitations and addressing high-value-added processes

Competitor Comparison

Technological Differentiation Against Competitors

	Overseas Companies	InnotionTech
Equipment Scale	Centered on small to medium-scale equipment	Capable of designing and manufacturing large-scale equipment
Applicable Products	Focused on small parts such as cutting tools, molds, and automotive parts	Broad application to large parts, batteries, semiconductor substrates, etc.
Process Temperature	400–500°C high-temperature process → Limited applicable materials, post-treatment required	Low-temperature (below 70°C) process → Applicable to various materials, minimized post-treatment
Coating Technology	PVD / PACVD (Mainly single methods)	Hybrid Plasma (PVD + CVD Fusion) → Capable of realizing high-functionality nano-composite films
Industrial Application	Focused on tools and general industrial parts	Expandable to high-value-added industries such as batteries, semiconductors, displays, and bio
Listing Status	Listed on SIX Swiss Exchange (Market cap approx. USD 1.19 Billion)	Unlisted

CONFIDENTIAL

Innovative Technologies with Strong Barriers to Entry

01 Hybrid Dry Coating Combining PVD + CVD



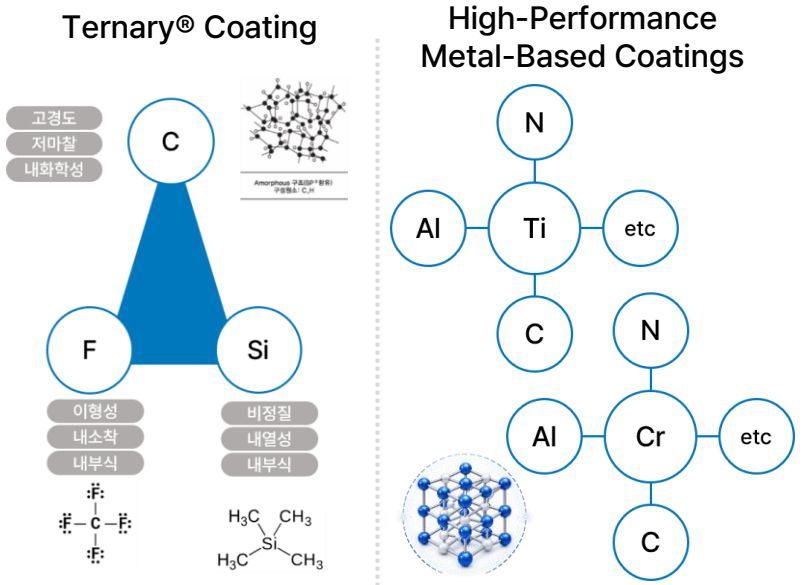
- Applicable to a wide variety of materials
- Capable of high-functionality thin-film coating
- Reduced coating process time compared to similar technologies

02 In-House Design, Manufacturing & Automation of Coating Equipment



- Capable of self-production
- Possession of large-scale equipment applicable to coating large products

03 Accumulated Coating Materials and Process Expertise



- Possession of diverse coating know-how
- Rich experience with various process recipes
  Providing customized solutions tailored to product characteristics
- Proven track record verified in industrial fields

CONFIDENTIAL

Hybrid Dry Plasma Coating

Application of high-functionality thin-film deposition technology in single-layer form

Nano-composite single-layer structure

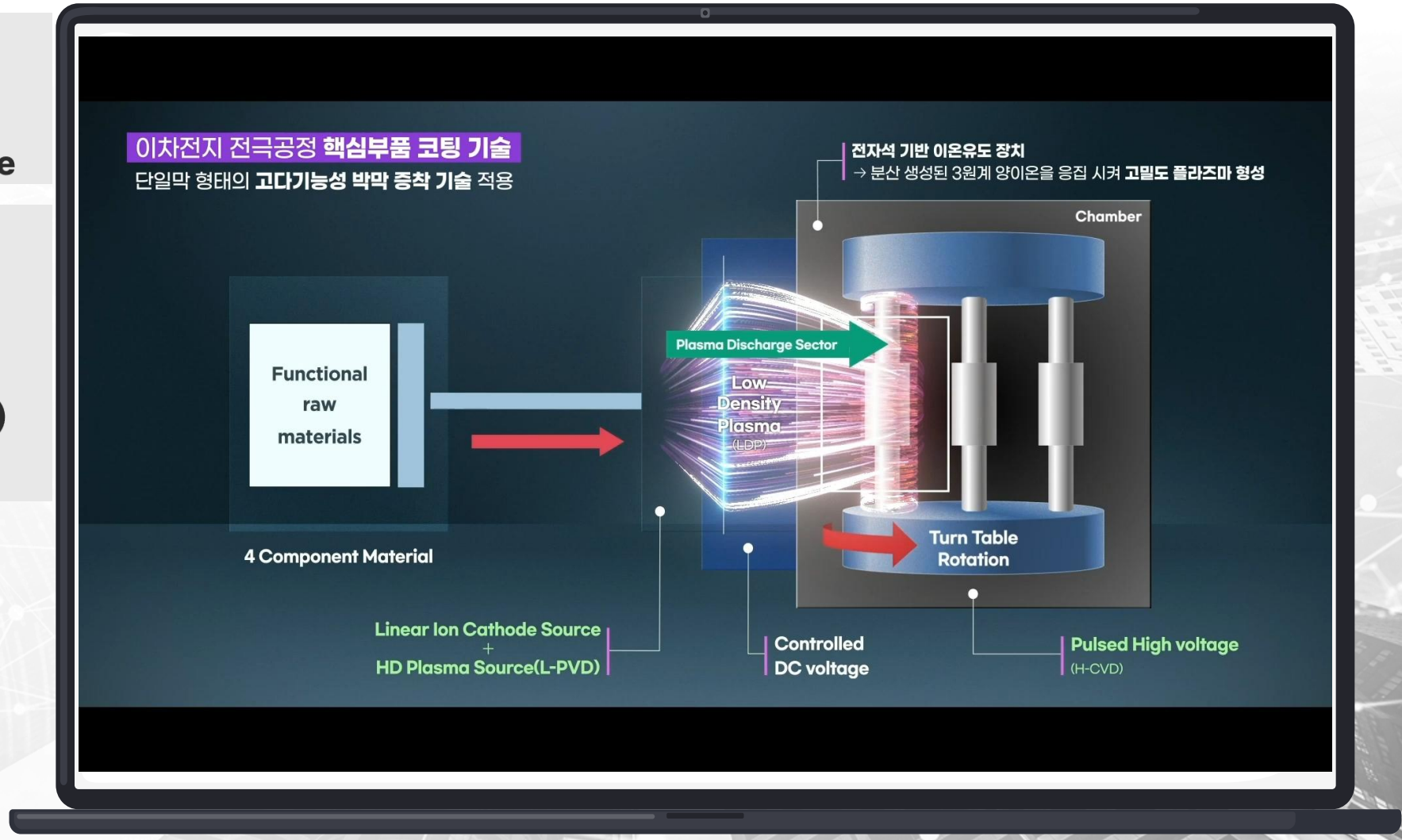
Simultaneous doping type utilizing ion induction devices
Application of single-layer nano-composite material deposition technology

Generation of High-Density Plasma (HDP) of target compositions

▼

Realization of high-performance and high-density thin films in a single layer

Simultaneous effect of process simplification



CONFIDENTIAL

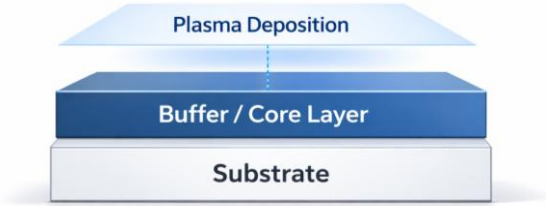
Hybrid Plasma Coating Process

Step 1. Activation & Cleaning → **Step 2. Adhesion Layer** → **Step 3. Top Layer**



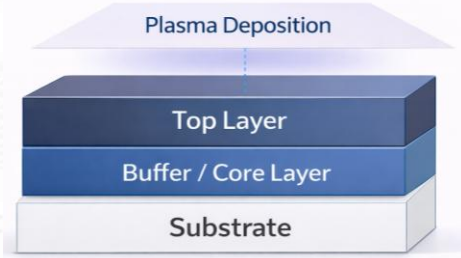
Step 1. Activation & Cleaning

- Ion Cleaning & Surface Activation (Enhanced Adhesion)
- Removal of contaminants from theproduct/ component surface and surface activation to improve adhesion between the substrate and coating layer



Step 2. Adhesion Layer

- Buffer Layer Coating (Enhanced Adhesion & Durability)
- Functional metal coating using a sputtering source for stress control
- Improves coating-to-substrate adhesion and overall durability

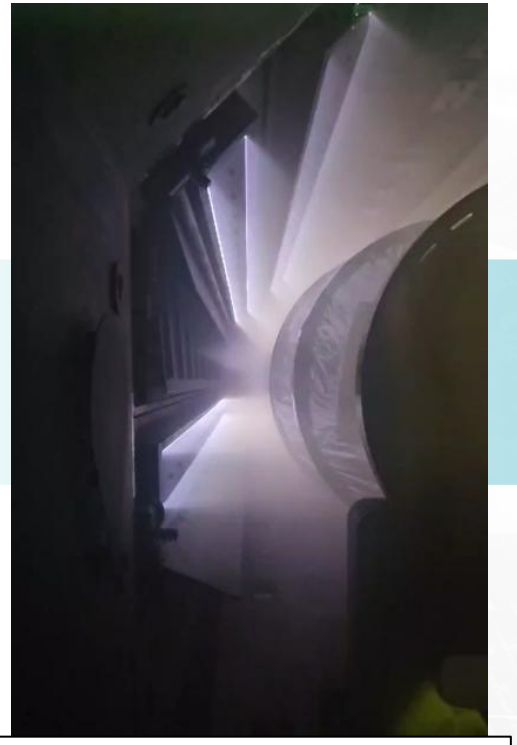


Step 3. Top Layer

- Functional Top-Layer Coating
- Formation of a high-density single coating layer through a hybrid plasma discharge process incorporating ion and magnetic-field control technologies

CONFIDENTIAL

Hybrid Plasma Coating Process



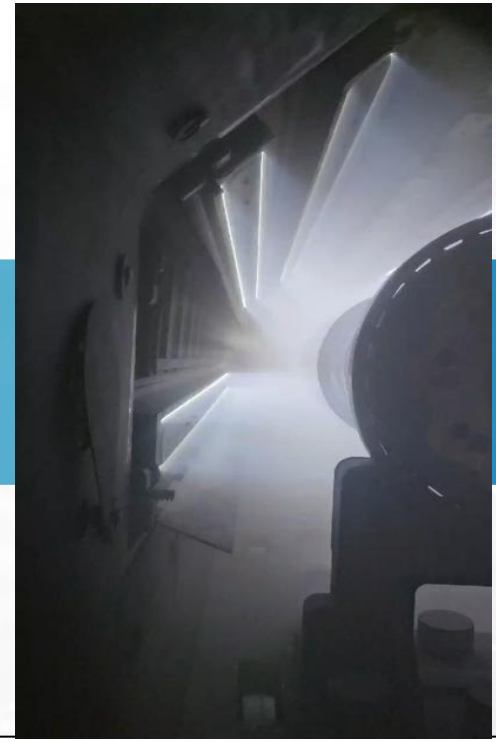
1 Step

(Ion cleaning & Surface Activation)



2 Step

(Hard Metal Base Nano-multi layer)



3 Step

(Functional & Top layer)

1cycle time : about 9hour_ (Capa: Wet battery Roll 1EA/batch , Dry Battery Roll 4EA/batch)

CONFIDENTIAL

Large-scale component coating equipment : From design to automation control

Application of coating technology for medium-to-large (~up to 10 tons) components with 5M-class chambers

: 100% Innotiontech in-house manufacturing process

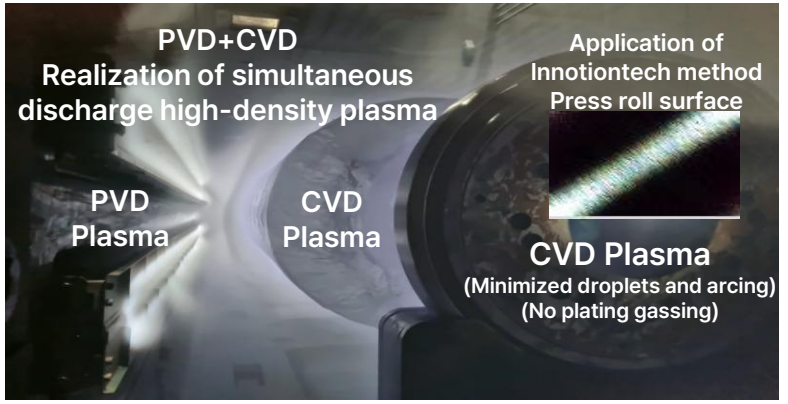
Innotiontech L-HPCS equipment image



Siheung HQ
Facility & Equipment

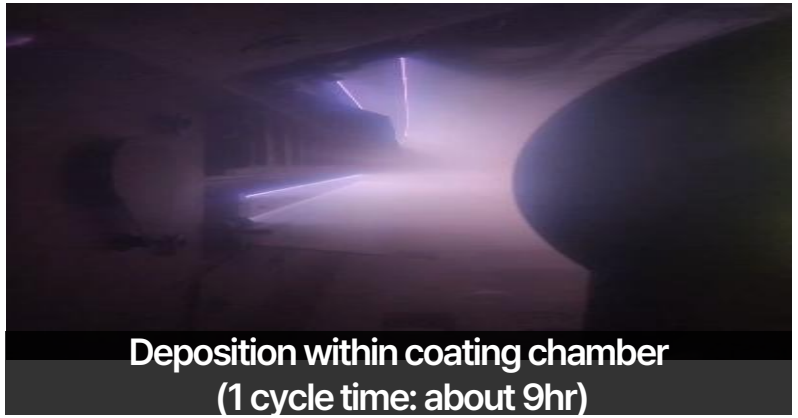


Busan Branch
Facility & Equipment





Before coating (Hard Cr plating)



Deposition within coating chamber
(1 cycle time: about 9hr)



After coating (Hard Cr plating + Innotion coating)

CONFIDENTIAL

Technology Built on Years of Accumulated Process Recipes and Optimization Data

Overcoming the Limitations of Conventional Coating Technologies



Limitations of Single-Layer Metallic and Hydrocarbon-Based Coatings

- High-temperature processes (>70°C) and thermal deformation
- Droplet defects and limited applicability to complex geometries

Next-Generation Coatings Based on High-Performance Nanocomposite Materials



Synergy of Nanocomposite Coating Materials and Hybrid Processes

- Ultra-low-temperature processing (<70°C) and highly dense thin-film formation
- Highly adherent thin films with near-zero droplets and pinholes

Proprietary Material Recipes and Optimized Coating Solutions



Customized Solutions Optimized for Diverse Materials and Component Environments

- Customized recipe design for a wide range of materials and components
- Extreme performance and next-generation specifications enabled by validated nanocomposite recipes

Beyond Technology: Years of Accumulated Recipes, Know-how, and Experience Deliver Optimized Solutions

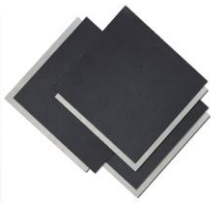
CONFIDENTIAL

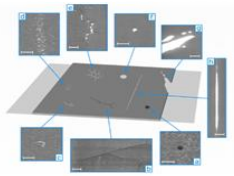
Coating Technology Demonstrated through Ternary®

Secondary battery manufacturing process essential technology (Problems)

1. Wear, chipping, corrosion, and material adhesion on production equipment components
   → **Reduced productivity and increased defects in electrode manufacturing**
2. Increased maintenance costs due to short component service life
   → **Higher manufacturing costs**
3. Electrode quality issues caused by degraded calender roll performance
   → **Reduced mass-production yield and battery performance**







Applying multifunctional nanocomposite coating technologies optimized for individual components, based on accumulated coating material expertise and process analysis capabilities

Overcoming the limitations of conventional coating materials through the **development of high-performance ternary nanocomposite coatings**

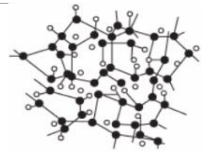
Ternary® (3C) Nanocomposite Properties

(Core Patents Granted in Korea and Overseas)

Base Material

High hardness
Low friction
Chemical resistance

C

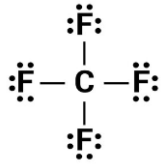


F

Si

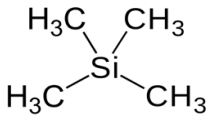
Add Low friction Functional Material

Release properties
Anti-seizing
Corrosion resistance



Nanocomposite Driving Minor Material

Heat resistance
Amorphous
Corrosion resistance

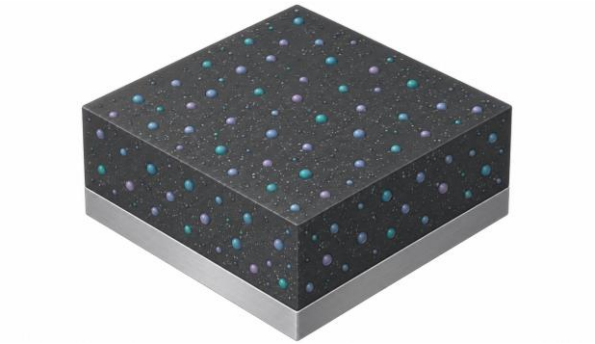


Realizing high-functionality thin-film characteristics through single-layer synthesis based on quaternary systems

CONFIDENTIAL

Advantages of Ternary® Coating

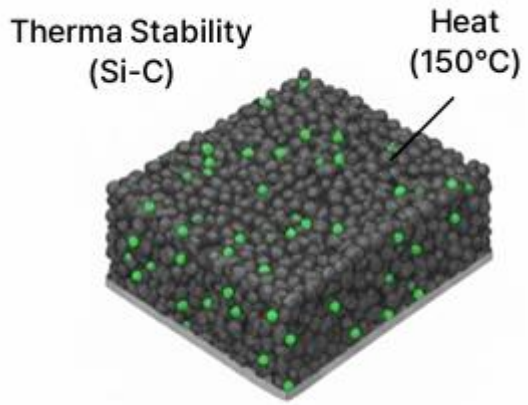
Nano-Composite Structure



Key Feature: Multi-Functional Synergy

- Integration of different functional elements within a nano-scale structure
- Balanced performance in friction, thermal stability, and surface properties, rather than maximizing a single property
- Tailored surface properties through composition and microstructure control

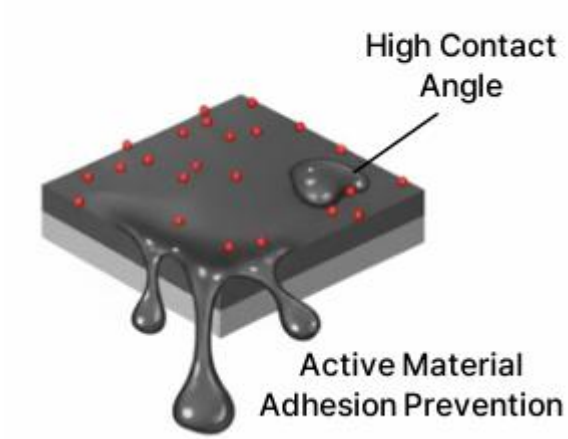
Functional Element: Si (Silicon)



Key Feature: Thermal Stability

- Strengthens the carbon network structure
- Suppresses structural degradation and phase separation under high-temperature conditions

Functional Element: F (Fluorine)



Key Feature: Anti-Sticking

- Significantly reduces surface energy through fluorine bonding
- Prevents active materials from adhering to the surface

CONFIDENTIAL

Advantages of Ternary® Coating

Category		DLC	Ternary®
	Coating Composition	C, H	C, H, Si, F(H)
	Heat Resistance	~ 250°C	~ 350°C
	Adhesion	HF1 ~ HF3	HF1 ~ HF3
	Coefficient of Friciton	0.10~0.15	0.05~0.10
	Hardness	15~23GPa (1,400 ~ 2,200 HV)	15~23GPa (1,400 ~ 2,200 HV)
	Contact Angle	50 ~ 60°	80 ~ 90°
	Active Material Adhesion	Occurrence	Suppressed / Prevented
	Durability test (110°C)	≥8,000km	≥15,000km
	Roll Process Temperature	Cold Process	Cold & Hot Processes
	Applicable Electrode Process	Wet Process	Wet & Dry Processes

Ternary® is optimized for high-temperature calendering processes up to approximately 150°C.

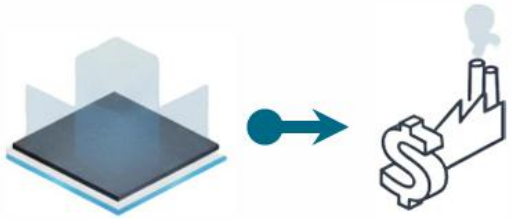
CONFIDENTIAL

Advantages of Ternary® Coating

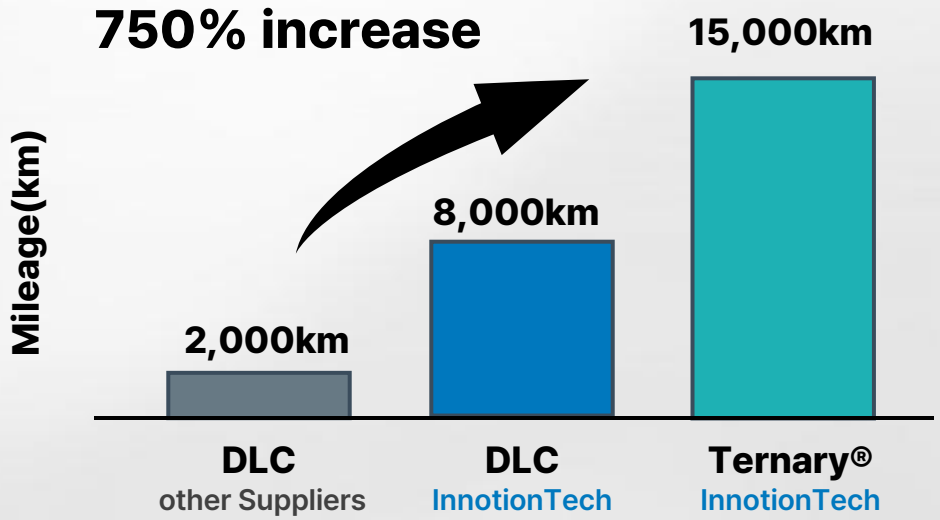
Key Technical Specifications

- High Hardness
- High Durability
- Wear Resistance
- Low Firction
- Thermal Stability
- Anti-Adhesion / Non-stick
- Surface Control
- Electrode Uniformity
- High-Density Thin Film
- Process Compatibility

Proven Performance Across Wet/Dry Electrode Processes and Cold/Hot Operating Environments



Expected Benefits in High-Temperature Operation



+α

Increased Production Uptime
- Reduced roll replacement and refurbishment frequency, increasing production uptime

Improved Quality & Yield
- Improved electrode performance and process yield

CONFIDENTIAL

Innotiontech Core Business Overview

Plasma Coating Technologies for Critical Components Across Advanced Industries — from Secondary Batteries to Semiconductors, Mobility, and Displays

Core Business

MAIN BUSINESS

Secondary Batteries

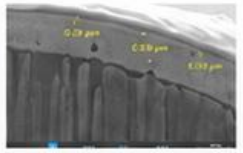




- Key Technology: Carbon-Based Nanocomposite Coatings
- Key Feature: High Hardness, Low Friction, Wear & Corrosion Resistance
- Applications: Transfer Equipment Parts, Glass-Contact Components

Semiconductor





- Tech: Functional Nanocomposite Coatings
- Features: Conductive Insulating, Low-Friction, Wear-Resistant
- Applications: Semiconductor Inspection Probes

Mobility





- Tech: $ZrSiO_2$, Nanocomposite Coatings
- Features: Low Friction, Wear & Electrical Erosion Resistance
- Application: EV Motor Bearings, Bipolar Plates

Display

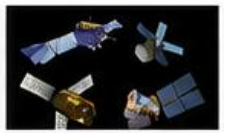




- Tech: Carbon Nanocomposite Coatings
- Features: High Hardness, Low Friction, Wear & Corrosion Resistance
- Applications: Transfer Parts, Glass-Contact Parts

Advanced Industries·Robotics





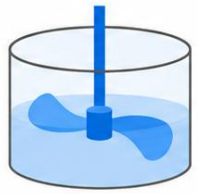
- Tech: Carbon Nanocomposite Coatings
- Features: High Hardness, Low Friction, Wear & HeatResistance
- Application: Solid-State Batteries, Aerospace, Robot Reducers

CONFIDENTIAL

In Wet Electrode Processing, Roller Surface Performance Drives Productivity

Roller Surface Treatment: A Key Technology for Electrode Quality and Productivity

STEP 01



Mixing process

Mixing active materials, conductive agents, and binder + solvent

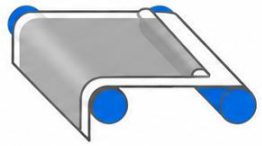
STEP 02



Slurry manufacturing

Manufacturing liquid-type slurry

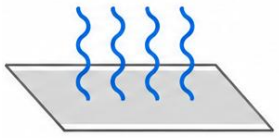
STEP 03



Slurry coating

Thinly coating cathode and anode slurries onto current collectors (aluminum foil, copper foil) respectively

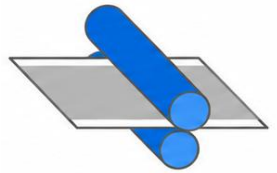
STEP 04



Solvent drying

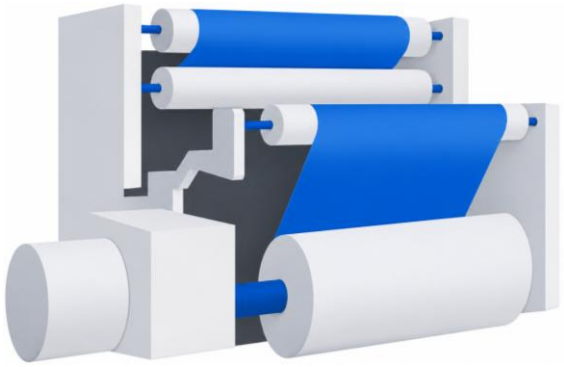
Drying in an oven at 100°C or higher

STEP 05



Roll pressing

Making electrodes thin and flat



Roll pressing process

- Rollers are core components that control the thickness and density of electrode sheets in the battery electrode manufacturing process
- Roller surface hardness, wear resistance, and release properties under repeated load environments are key determinants of electrode quality
- Roller surface treatment technology determines productivity and yield in the electrode manufacturing line

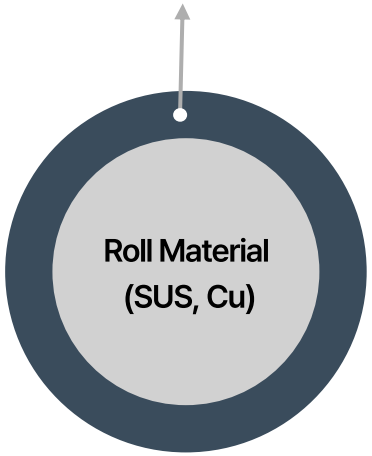
CONFIDENTIAL

Limitations of Conventional Wet Plating: Reduced Press Roll Durability

Limitations of conventional wet plating reduce press roll durability, directly impacting electrode quality, manufacturing cost, and safety.

1 Conventional Wet-Plated Industrial Roll Structure

Wet Cr Plating Layer: ~150 μm



- Limited durability due to lower coating density in liquid-based plating ↓
- Higher costs from raw materials and waste treatment ↑
- Conventional Wear Resistance (Surface Hardness: 900 HV)

2 Insufficient Durability of Electrode Press Rolls



[Wear of Hard Cr Plating
(within approx. 1.5 million m of use)]

Higher Manufacturing Costs for Battery Producers
Reduced Throughput & Electrode Quality Issues

3 Roller Surface Coating: Critical to Electrode Quality and Safety



Higher EV Costs
Increased Risk of Electrode Defects and Thermal Safety Issues

CONFIDENTIAL

Advanced Roller Surface Treatment Technology Beyond Conventional Wet Plating



HCr Plating
(Before Coating)



Innotion Coating
(After Coating)

Expected Benefits

1 High hardness · Low Friction

High hardness(17~25GPa) & low coefficient of friction (≤0.15)

→ Improved wear resistance and easier roll surface cleaning /maintenance

→ Reduced foreign-particle contamination through minimized wear

→ Extended maintenance intervals, improving productivity and reducing maintenance costs

2 Uniform Surface Roughness

Maintains consistent roll surface roughness through amorphous coating

→ Stabilizes surface quality during electrode production

3 Improved Release Property

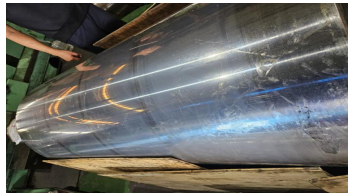
Surface property control through optimized F–Si–C:H composition

→ Reduces adhesion of active materials and Al/Cu electrode materials

→ Reduces roll wear through low-friction characteristics

CONFIDENTIAL

Surface Treatment Types & Characteristics in Wet Processes

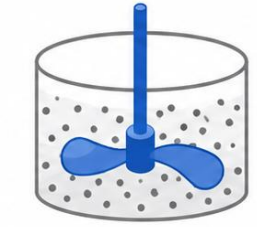
	HCr Plating / Polishing	HCr Plating+ INT Hybrid DLC	HCr Plating+ INT Hybrid Ternary®
Load	2~4 ton	6~7 ton	6~7 ton
Speed	60m / min	120m / min	120m / min
Lifetime	~1.5 million m / 1 month	12 million m / 3.5 months	~20 million m / 5 months
Quality & Evaluation	Short service life due to wear Not reusable when switching electrode models (Low productivity)	5–7 million m Maintains Grade-A initial electrode quality (Roll replacement due to maintenance of other equipment parts)	5–7 million m Maintains Grade-A initial electrode quality (Roll replacement due to maintenance of other equipment parts)
Wet Process Cvaluation			

CONFIDENTIAL

Dry Electrode Process: More Demanding Operating Conditions

Roller surface treatment becomes even more critical under demanding conditions requiring different surface functionalities.

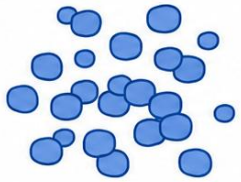
STEP 01



Mixing process

Mixing active materials, conductive agents, and binder

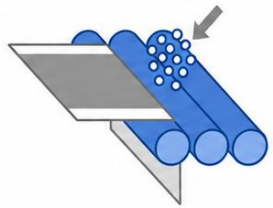
STEP 02



Solid Powder Preparation

Preparation of solid-state powder

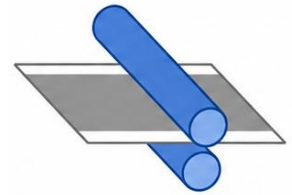
STEP 03



Dry Powder Coating

Coating solid powder onto the current collector

STEP 04



Calendering

Compressing the electrode to achieve the required thickness and uniformity



Dry Powder Coating & Calendering Process

- Unlike wet processes, dry electrode processing uses solid powder, requiring higher roll surface performance.
- Both press rolls (PS rolls) and film-forming rolls are required, with each roll demanding different surface functionalities.
- As roll types diversify and operating conditions become more demanding, advanced surface treatment technology becomes increasingly critical

CONFIDENTIAL

Limitations of Conventional Surface Treatments (PS Roll & Film-Forming Roll)

Dry electrode processing requires both PS rolls and film-forming rolls, each with different surface property requirements.

1

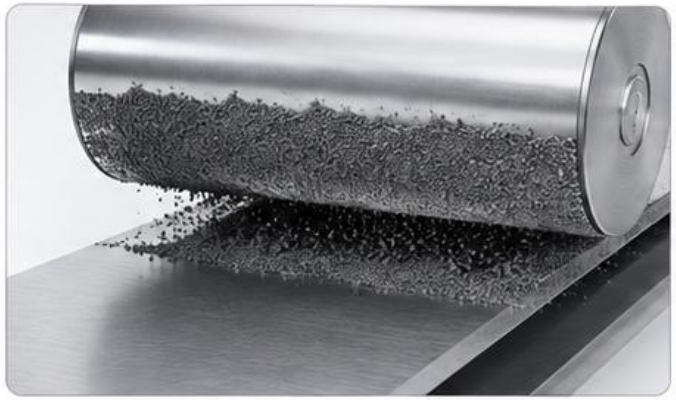
PS Roll : More Demanding Forming Conditions than Wet Processes



- Direct compression of dry powder under simultaneous thermal and mechanical loads
- Surface non-uniformity can cause variations in electrode thickness and quality
- Required Properties: Surface Uniformity · Release Property · Thermal Stability

2

Film-Forming Roll: Insufficient Adhesion → Poor Transfer



- The electrode must adhere stably to the roll surface
- Unstable transfer to the next roll can interrupt electrode production
- Required Property: Controlled Electrode Adhesion

3

Film-Forming Roll: Speed Mismatch → Accelerated Wear



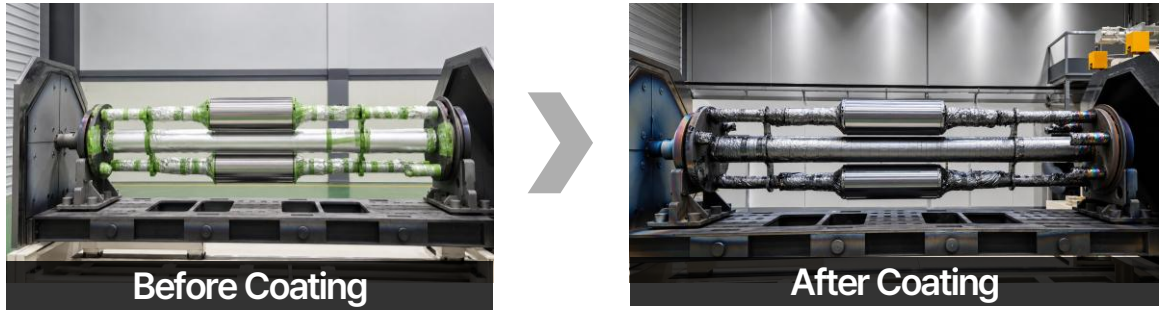
- Mismatch between roll rotational speed and sheet transport speed
- Continuous friction accelerates surface wear and reduces service life
- Required Properties: High Hardness · Wear Resistance

CONFIDENTIAL

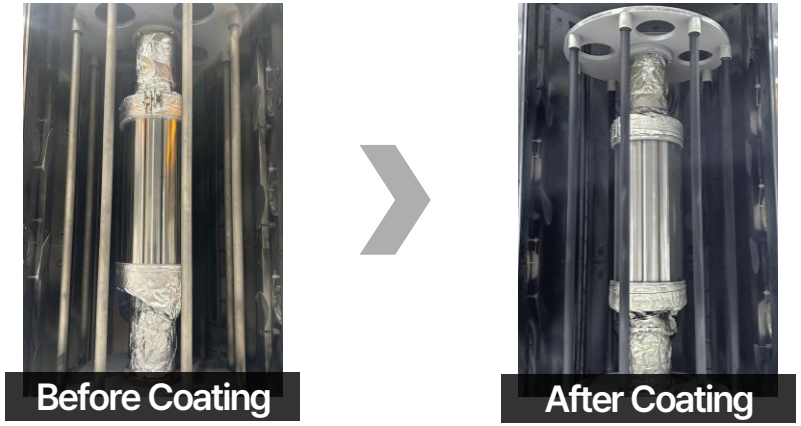
Roller Surface Treatment Optimized for Dry Electrode Processing

Dual-Material Design Tailored to the Required Properties of PS and Film-Forming Rolls

PS Roll (Ternary®)



Film-Forming (Metal-Based Coating)



Expected Benefits

1 High-Temperature Dry Process Capability (PS Roll / Ternary®)

Ternary nanocomposite coating with heat resistance up to ~350°C

→ Ensures thermal stability during direct compression of dry powder

→ Reduces sticking and improves sheet quality consistency

2 Controlled Electrode Adhesion (Film-Forming Roll / Metal-Based Coating)

Metal-based surface design for stable electrode transfer

→ Maintains surface uniformity during electrode production

→ Enables stable electrode transport through controlled transfer behavior

3 High Hardness & Wear Resistance (Film-Forming Roll / Metal-Based Coating)

Resists slip-induced wear caused by speed differences between process sections

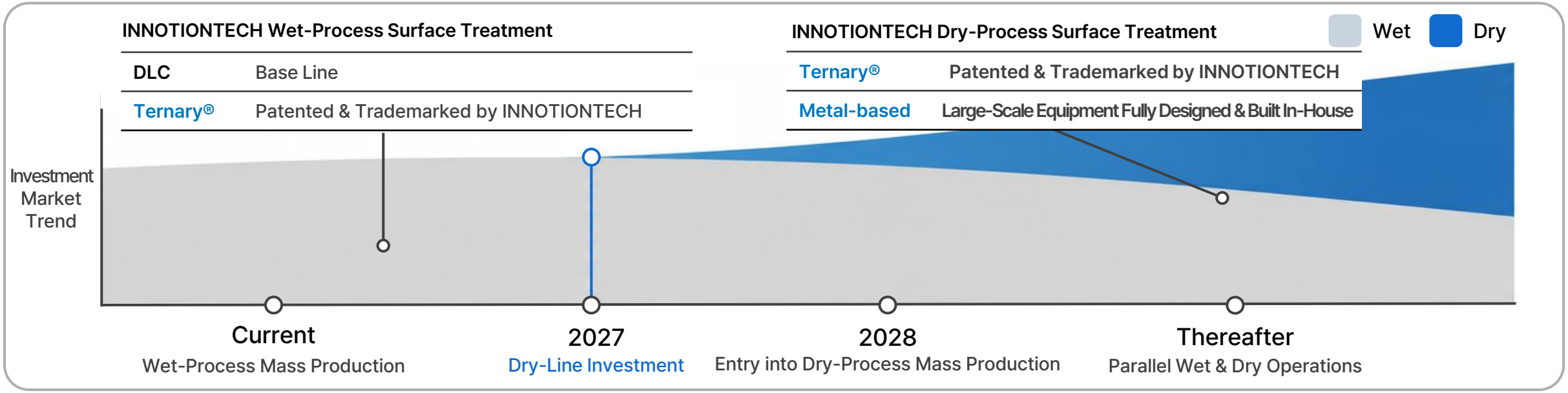
→ Extends roll service life and maintenance intervals

→ Dense coating structure without pore-induced outgassing compared with WC thermal spray

CONFIDENTIAL

Investment Trends in Secondary Battery Electrode Processing

Equipment replacement does not happen all at once — Dry-process revenue is added on top of the existing Wet-process revenue base



Replacement Proceeds Gradually, Starting with Aging Equipment

Production lines are not replaced all at once
Aging equipment is replaced sequentially, allowing Wet-process lines to remain in operation for an extended period

Adoption Rates Vary by Customer

Customers will not transition to dry processing at the same time.
LGES is leading the transition, while other customers remain at earlier stages, sustaining demand for Wet-process solutions.

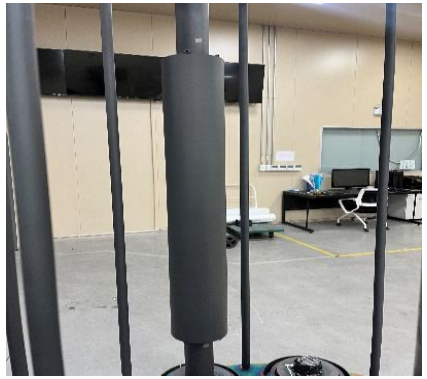
Existing Wet-Process Assets Carry Over into Dry

Existing customer relationships, vendor positions, and global infrastructure can be leveraged directly as customers transition to dry processing.

CONFIDENTIAL

Expansion Across Key Electrode Process Stages

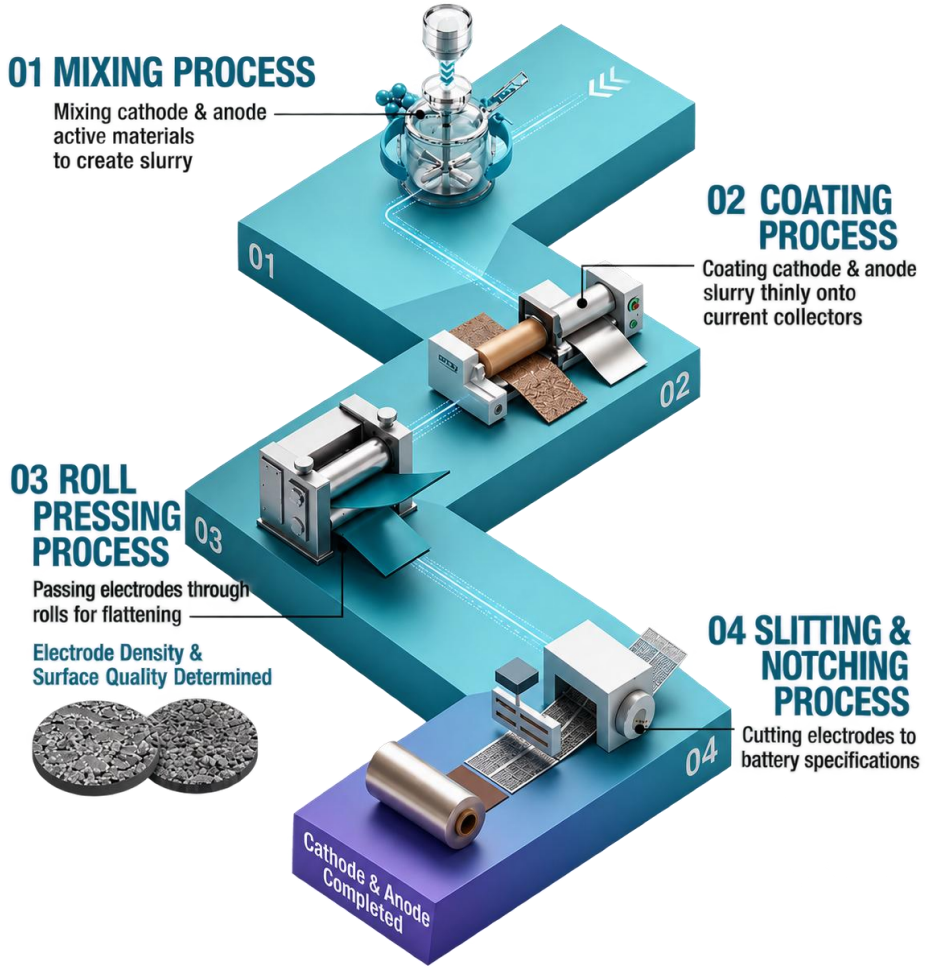
Expanding coating application areas centered on the 'Press Roll (PS Roll)' process



- Rubber Roll Coating for Cathode Material Equipment
- Order Received for 4 Units for HLI



- Press Rolls / Pattern Rolls / Guide Rolls for Electrode Processing
- LGES Coating Specification Finalized & Vendor Designated





- Coating for Electrode Slot-Die Nozzle Components
- Currently Supplying to a Listed Company (GITEC)



- Slitter Knife Coating

CONFIDENTIAL

Investment Execution & Mass-Production Infrastructure for the Secondary Battery Business

Building Mass-Production Capabilities across Korea, the U.S. and Europe (5 Large-Scale / 7 Small & Mid-Scale Systems in Total)

2023 | 2024 | 2025 | 2026

HQ Mass-Production Site
(Company-Owned)

Mass-production facility established at Siheung HQ
Established domestic customer references
Large-Scale: 2 / Small & Mid-Scale: 5

Sales & Technical Support



Busan Production Site
(Leased)

Busan production facility established
Established mass-production capability for large rolls
Large-Scale: 2 / Small & Mid-Scale: 1

Large-Roll Mass Production



U.S. Subsidiary
(Leased)

Weirton Subsidiary Established
Completed local equipment setup in the U.S.
Large-Scale: 1 / Small & Mid-Scale: 1

Local Support for North America



H1

Relocation to Busan Facility
(Company-Owned)

Mass-Production Equipment & Operations
Large-Scale: 2 / Small & Mid-Scale: 1
Mid-to-Large Scale: 1

Supply Chain Management & Logistics Hub



H2

Czech Subsidiary in Europe
(Leased) Planned

Ostrava Site Planned
Targeting Mass Production in Feb. 2027
Large-Scale: 2 / Small & Mid-Scale: 1

Support for the Eastern European Cluster

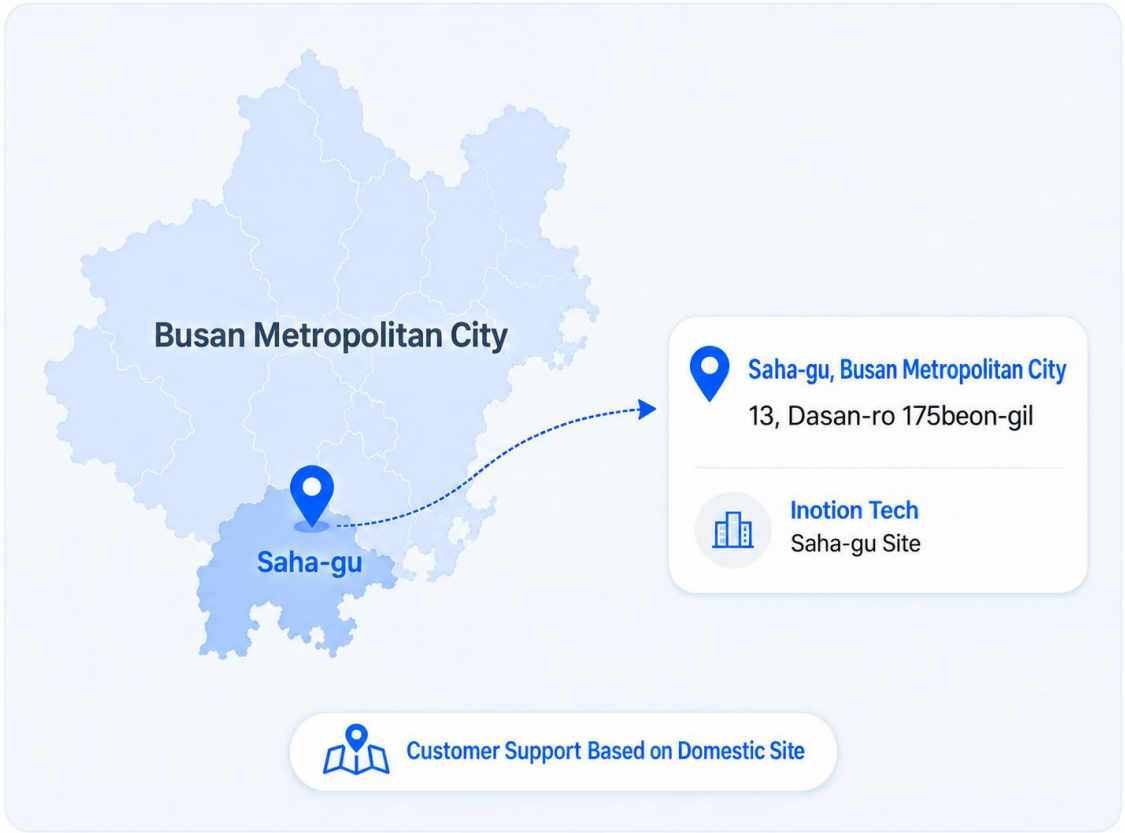


CONFIDENTIAL

INNOVATIVE PLASMA COATING INOTION TECH

Relocation of Busan Production Site

Strengthening Production Capacity to Support Domestic Battery Mass Production & Growing Global Demand



- Secured a company-owned production site through the relocation of the Busan branch and acquisition of land and facilities
- Expanded production infrastructure to support large rolls and mass-production demand from domestic battery customers
- Enhanced production stability and capacity to support future growth in global demand

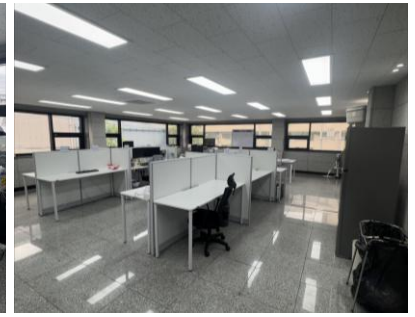
[Busan Production Site Overview]











CONFIDENTIAL

North American Production Base Fully Established

Production base for local customer response in North America



- INNOTIONTECH USA INC. established on September 10, 2025
- Shop-in-shop operation at NEO Industries' Weirton plant, specializing in plating and grinding services for battery press rolls in the U.S.
- Setup and commissioning of dedicated press roll coating equipment completed in January 2026
- Coating price negotiations and vendor registration underway for UCL2 (Tennessee) (through Oct. 2026)
- Accelerating market entry and expansion through collaboration with NEO to provide responsive local customer support in the U.S.

[Weirton Plant – Setup Completed]





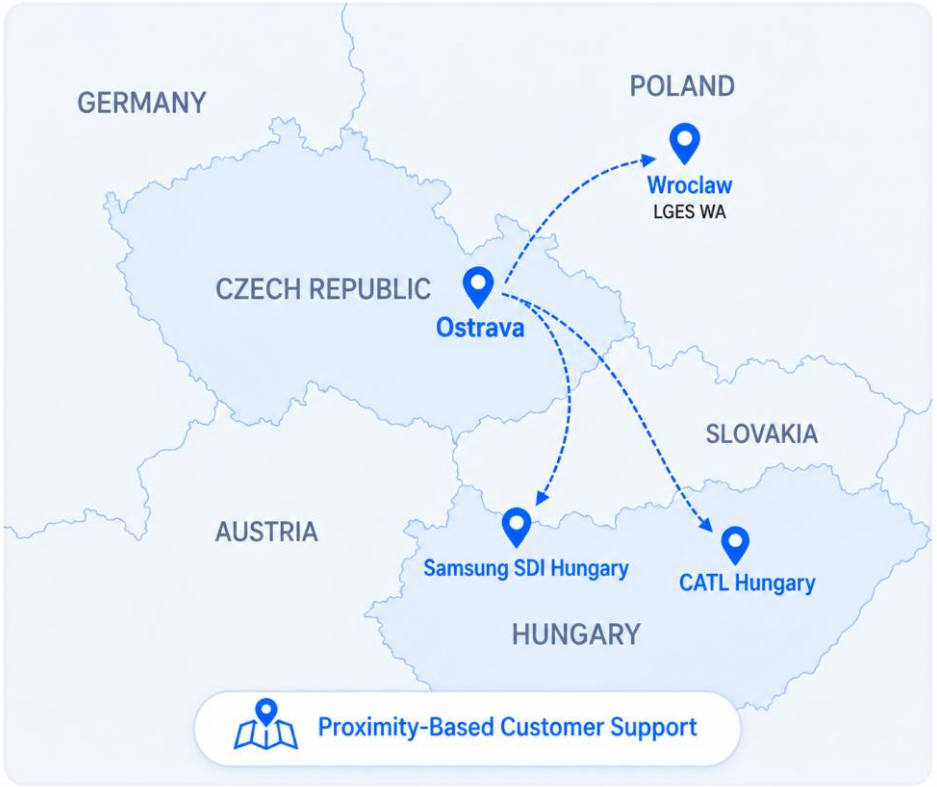


CONFIDENTIAL

INNOVATIVE PLASMA COATING INNOTION TECH

European Local Production Base Planned

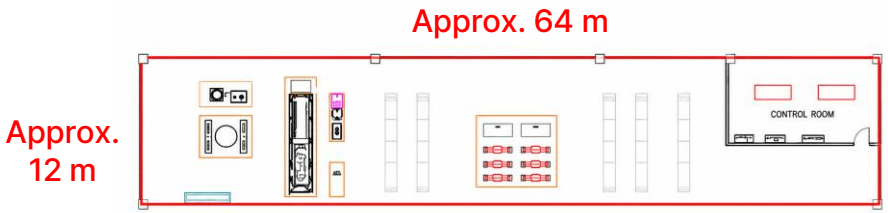
Establishment for local customer response in Europe

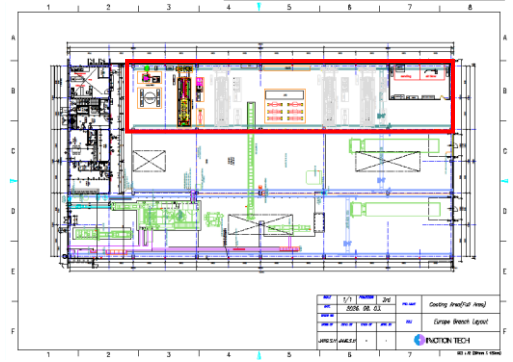


- INNOTIONTECH s.r.o. incorporated on September 2, 2026
- Shop-in-shop operation at NEO-JL Industries (CTPark Poruba, Ostrava, Czech Republic)
- Local facility construction underway (approx. 5 months), targeting mass production in H1 2027
- Approx. 767 m² production area + 46 m² office = 813 m² total

Incorporation Documents







[Ostrava Plant]





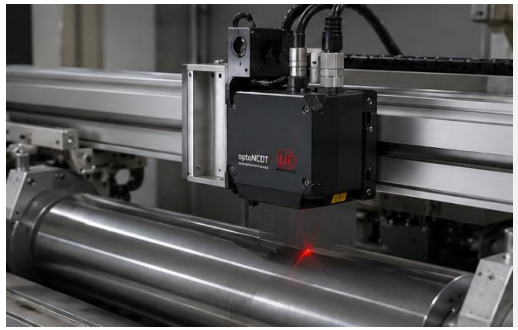


CONFIDENTIAL

New Business – Roll Surface Vision Inspection System

Automated Vision Inspection Solution for Roll Surfaces, Optimized for Electrode Manufacturing Environments





Electrode Press Roll
Detects various inspection items, including surface defects, foreign particles, pinholes, and surface uniformity

Real-Time AI Deep Learning Inspection

Inspects roll surfaces based on trained data and classifies defects in real time as OK or NG

Defect Location Tracking

Converts curved roll images into flat-view images to accurately track defect coordinates

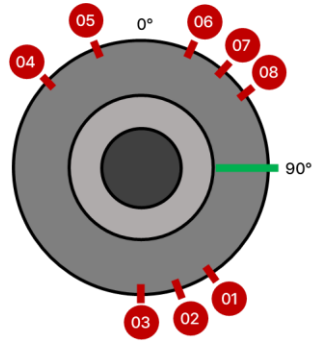
Data Classification & History Management

Displays results while saving inspection data by judgment category in real time and systematically managing history for statistical analysis

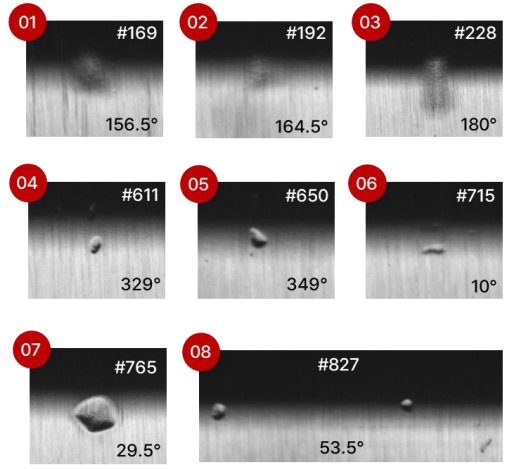
High-Precision Optics & Lighting Control

Precisely detects fine scratches and dents that are difficult to identify visually

Roll Cross-Section | 920 Circumferential Segments



Actual Defect Detection Images

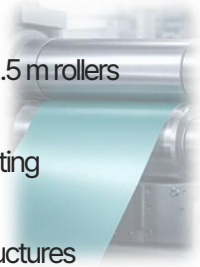


CONFIDENTIAL

Advanced R&D: Dry Cr Thick-Film Coating Technology

Limitations of Conventional HCr Wet Plating	Dry Plasma-Based High-Density Cr Thick-Film Technology	Superior Cost Efficiency & Market Potential
• Hexavalent chromium wastewater → Environmental & ESG risk • Thickness variation & porosity → Limited quality reliability Underlayer defects → Defect propagation to upper films • WC spray alternative → Porosity / outgassing → delamination & shorter lifetime	• Eco-friendly dry process → Zero wastewater & hazardous emissions • Large-area deposition → Uniform 100 μm coating on 2.5 m rollers • High-energy PVD → Excellent adhesion / no film lifting • Compatible with DLC → Optimized hybrid coating structures	• Extended roller lifetime → Stable electrode quality • Process Integration → One-stop dry coating (Cr underlayer + functional film) • Market leadership → Driving the shift from wet plating to dry coating







Expandable Across Industries Replacing Chrome Plating (Global hard chrome plating market: approx. KRW 12T as of 2025)

Industrial Rollers Shafts	Automotive & Machinery Components	Semiconductor & Display Equipment Parts	Energy & Plant Equipment
• Paper, film, steel rolling rolls, printing rollers	• Piston rods, cylinders, mold surfaces	• Friction-, wear-, contamination-sensitive components	• Pump shafts, valves, rotating components

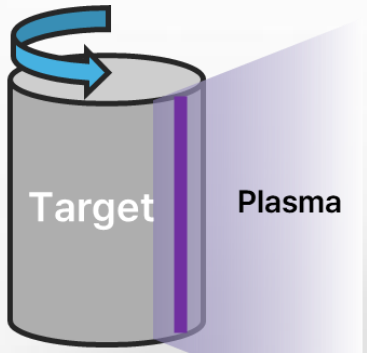
A next-generation surface treatment platform driving the "transition beyond Cr plating"

CONFIDENTIAL

Advanced R&D: Dry Cr Thick-Film Coating Technology

Large-Area Dual-Magnet Cylindrical Rotating High-Power, High-Cooling, High-Density Plasma System

Large-Area · High-Cooling · High-Power Sputtering System



- Direct cooling system compared with conventional planar targets

→ Enhanced cooling enables high-power operation

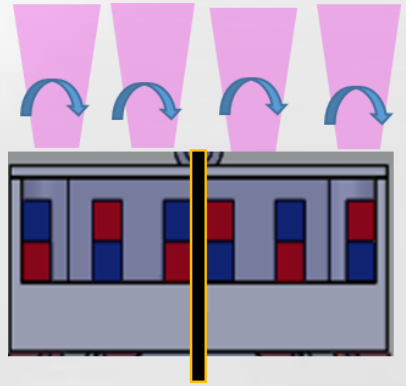
- Cylindrical rotating structure

→ More uniform target erosion and higher target utilization

- Minimized heat concentration even under high-power conditions
- Uniform deposition on large-area substrates up to 2,500 mm



Dual-Magnet-Based High-Density Plasma Control Technology



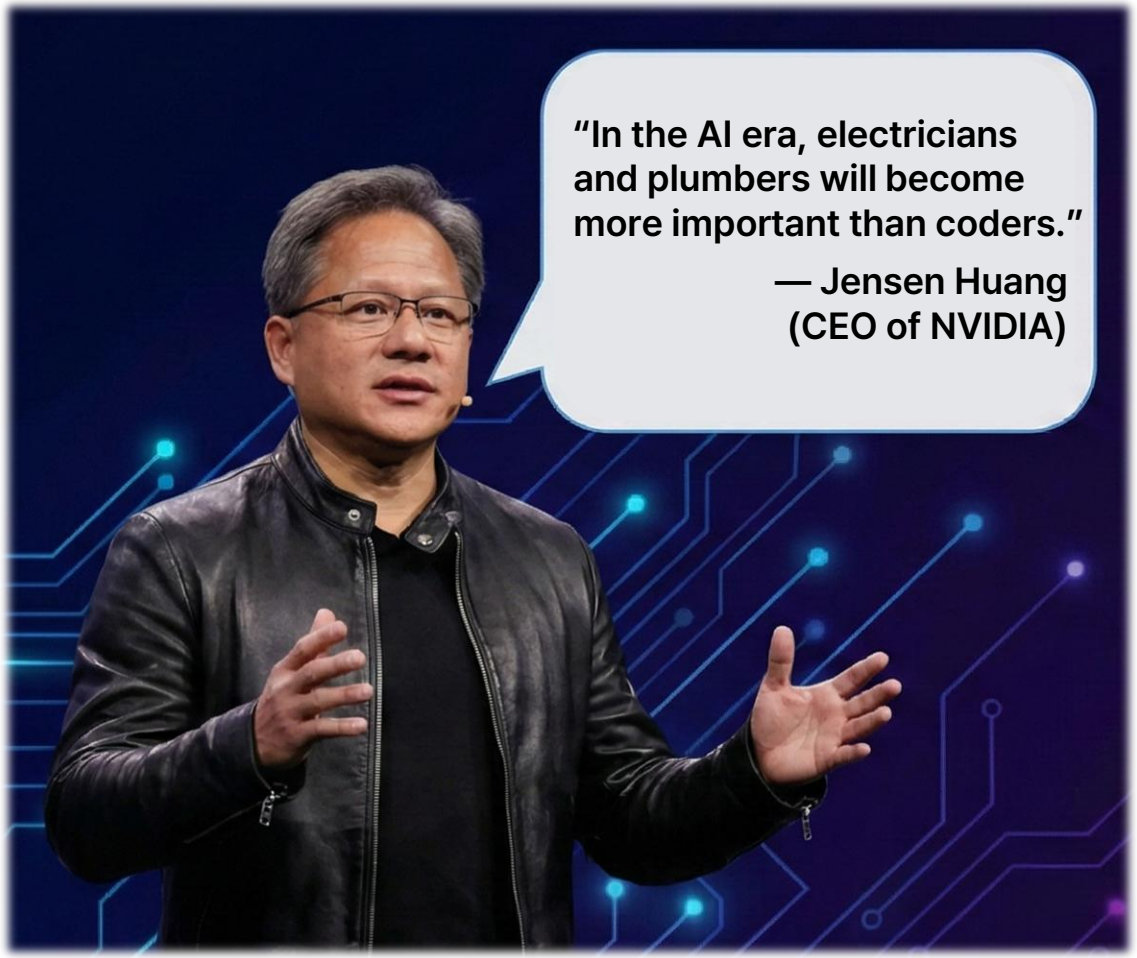
- In battery electrode manufacturing, rollers are key components that control electrode sheet thickness and density
- Under repeated loading, surface hardness, wear resistance, and release properties of rollers are critical factors determining electrode quality
- Roller surface treatment technology directly affects productivity and yield of electrode manufacturing lines

Enables dense thin- and thick-film coatings using high-power, high-density plasma technology for large-area rolls

CONFIDENTIAL

Strategic Insight: The Return of Foundational Technologies to Solve Bottlenecks in Advanced Industries

As technology advances, industrial success is determined not by software, but by the ability to overcome physical limitations.



INNOTIONTECH's Perspective

As AI becomes more advanced, the physical loads that AI-powered machines—such as robots, satellites, and servers—must withstand increase dramatically. The market now requires advanced foundational technologies capable of overcoming the limitations of conventional manufacturing processes.

Increasingly Harsh Operating Environments

- High-speed precision control in robots
- Extreme conditions in aerospace
- Ultra-high heat generation in data centers

Limitations of Conventional Technologies

- Inconsistent quality
- Limited capability for multi-functional performance
- Environmental regulation constraints of conventional processes

Surface Treatment as an Enabler

Innotiontech Hybrid Coating Solutions overcoming the limitations of conventional processes



CONFIDENTIAL

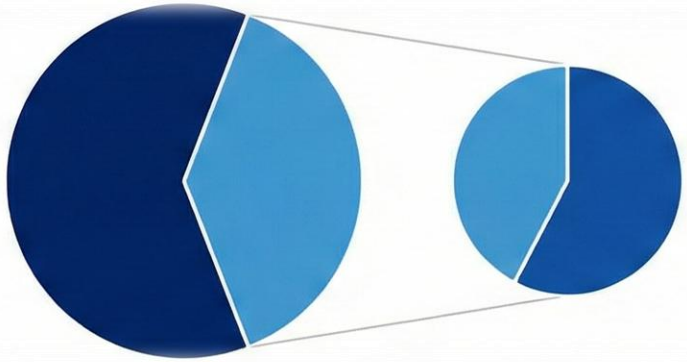
Market Expansion: Transforming a KRW 24T Global Legacy Market

Leading a new high-value market through the "Dry Transformation" of surface treatment technologies

Global Cr Plating Market

USD 18.36B

(~KRW 24T, 2026 Forecast)



Global Hard Cr Target Market

USD 96.6B

(~KRW 13T, 2026 Forecast)

Strategic Targets by Industry

Industry Sector	Environmental and Manufacturing Limitations	Innotiontech's Expansion Strategy
Robotics (Physical AI)	Heat generation and friction loss during high-speed operation	Registering our coating as the "Standard Spec" from the design stage through joint development with reducer and actuator manufacturers
Aerospace	Heat generation and friction loss during high-speed operation	Developing close R&D collaboration with global aerospace prime vendors using our U.S. subsidiary (WV) as a base. Establishing ourselves as an essential partner within the core aircraft/rocket value chain, serving as the "Final Piece" that overcomes the limitations of existing plating processes
AI Infrastructure (Data Center)	Corrosion and thermal management issues in high-density servers	ntegrating as a "Core Component Coating Solution" within finished products through MOUs and Proof of Concept (PoC) performance verification with global AI infrastructure manufacturers
Next-Generation Energy	Precision and durability in battery/hydrogen processes	Expanding partnerships to global top-tier battery and hydrogen equipment companies and proposing maintenance standardization based on the LGES success case

Major Clients

A partner trusted by global leading companies



Secondary Battery

LG 에너지솔루션 SAMSUNG 삼성SDI

pnt People & Technology CIS SK on TESLA Panasonic

Automotive | Aerospace

HYUNDAI SPACEX

HYUNDAI STEEL 현대제철

Display | Advanced Materials

SAMSUNG 삼성전기 SAMSUNG ELECTRO MECHANICS SAMSUNG DISPLAY LG Display

SK microworks

Semiconductor | Energy

SAMSUNG SK hynix HANA MATERIALS HYOSUNG HEAVY INDUSTRIES 효성중공업

CONFIDENTIAL

Core strategies to accelerate growth



Expansion of core upstream industries

- Expansion of global secondary battery cell maker client base
- Expectation of increased demand for our coatings due to supply shortages in the secondary battery industry and the investment expansion following the paradigm shift toward dry processes
- Accelerating growth by broadening the range of applicable components within the electrode process

Expansion of production bases and global market entry

- Completion of local subsidiary establishment to target the U.S. market
- Strengthening local sales synergy through collaboration with U.S. partner Neo Industries → Rapid market penetration
- Planned expansion into Europe based on similar future strategies

Industry diversification and new product development

- Expansion from secondary batteries to high-value-added industries such as aerospace, future mobility, and semiconductors
- Revenue diversification strategy encompassing coating equipment sales
- Positioning as a global top player by utilizing market-disrupting technologies and business models, such as thick-film coating technology and shop-in-shop strategies

INNOVATIVE PLASMA C OATING
INNOTION TECH

With the power of foundational technology, we turn our customers' visions into reality.

